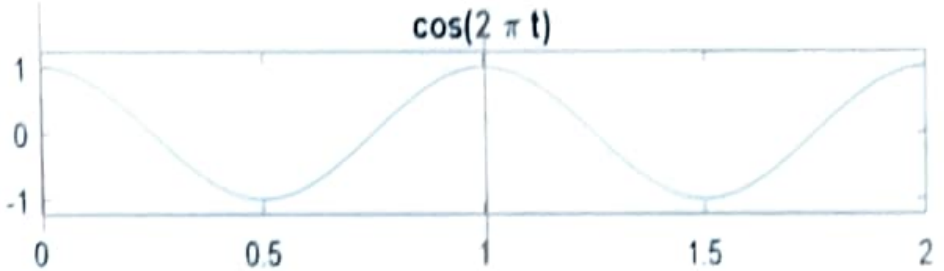
Sinusoidal signals

- $x(t) = A\cos(\omega t + \Phi)$
- A is the maximum amplitude of the sinusoidal signal
- ω is the radian frequency (**radians/sec**)
- Φ is the phase shift (**radians**)
- $\omega = 2\pi f$
- f is the frequency (**Hertz, cycles/sec**)
- $T = 1/f$
- T is the period (**sec**)



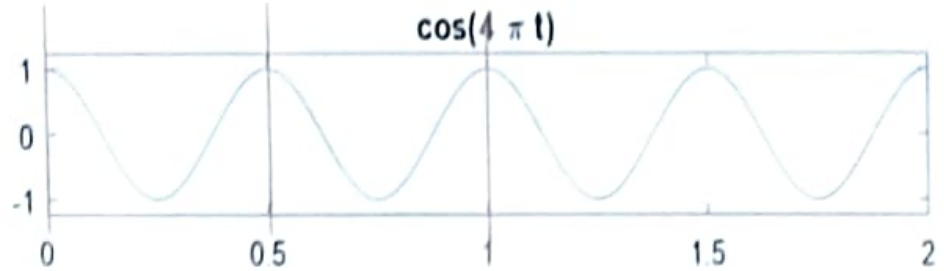
$$x(t) = A\cos(\omega t + \Phi)$$



$T=1$ sec

$f=1/T=$ 1 Hz

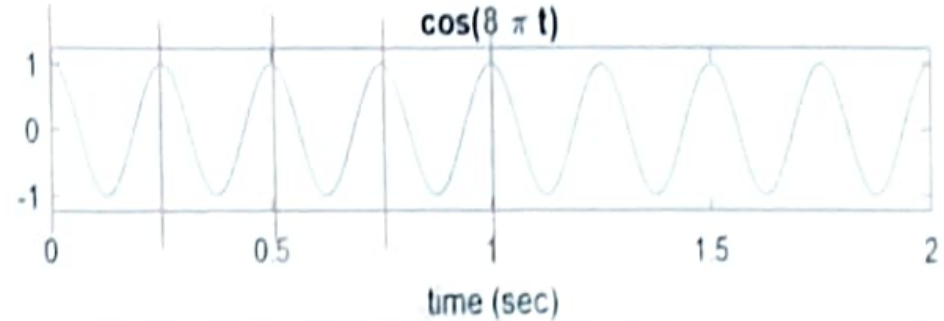
$\omega = 2\pi f = 2\pi$ rad/sec



$T=0.5$ sec

$f=1/T=$ 2 Hz

$\omega = 2\pi f = 4\pi$ rad/sec

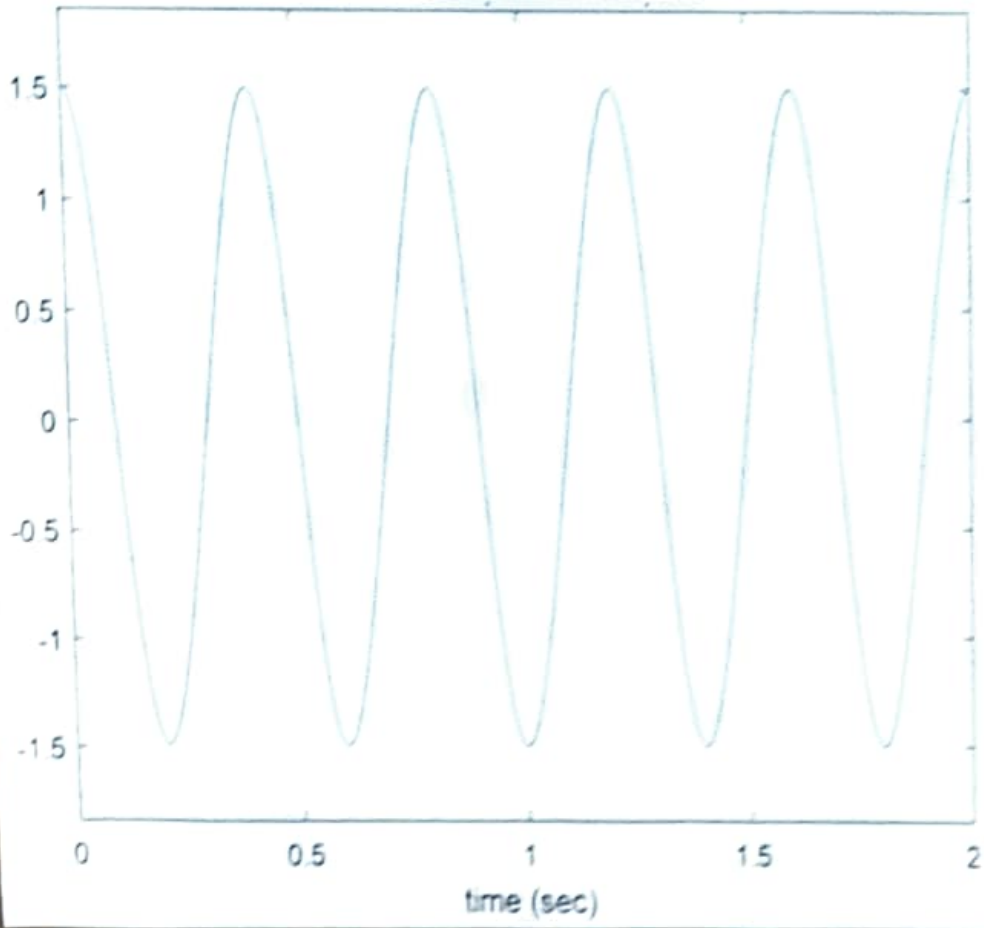


$T=0.25$ sec

$f=1/T=$ 4 Hz

$\omega = 2\pi f = 8\pi$ rad/sec

$x(t) = A\cos(\omega t + \Phi)$



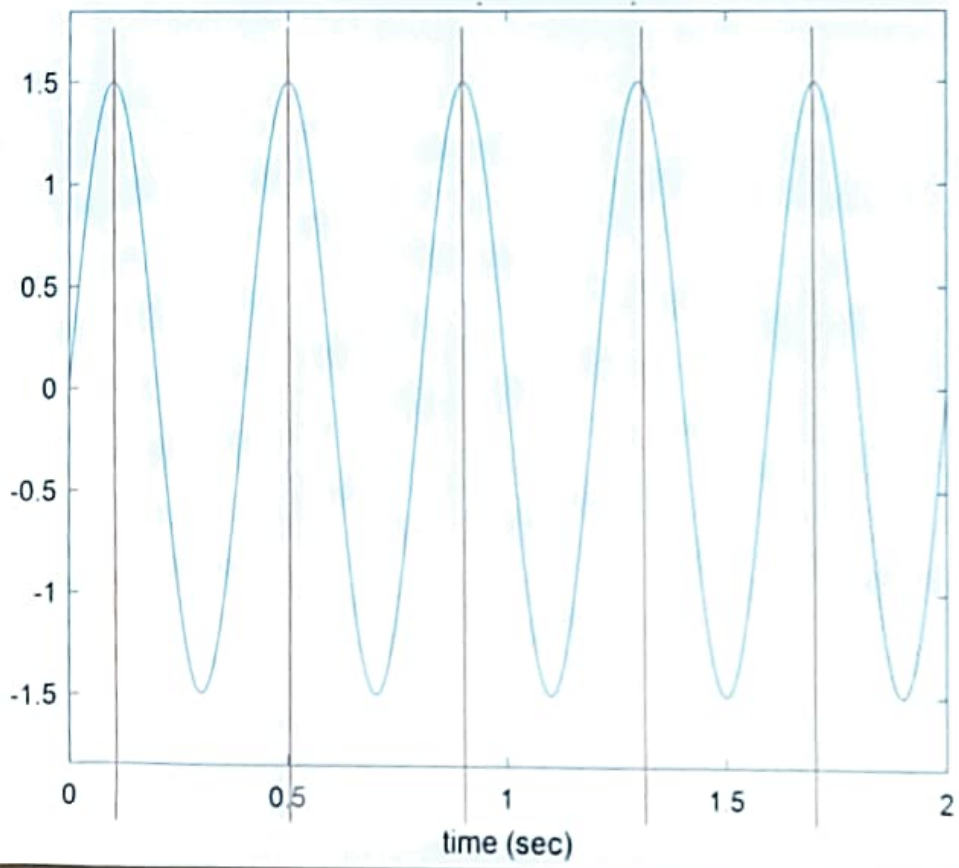
A=?

T=? sec

$f=1/T$= ? Hz

$\omega = 2\pi f$ = ? rad/sec

$x(t) = A\cos(\omega t + \Phi)$



$A=1.5$

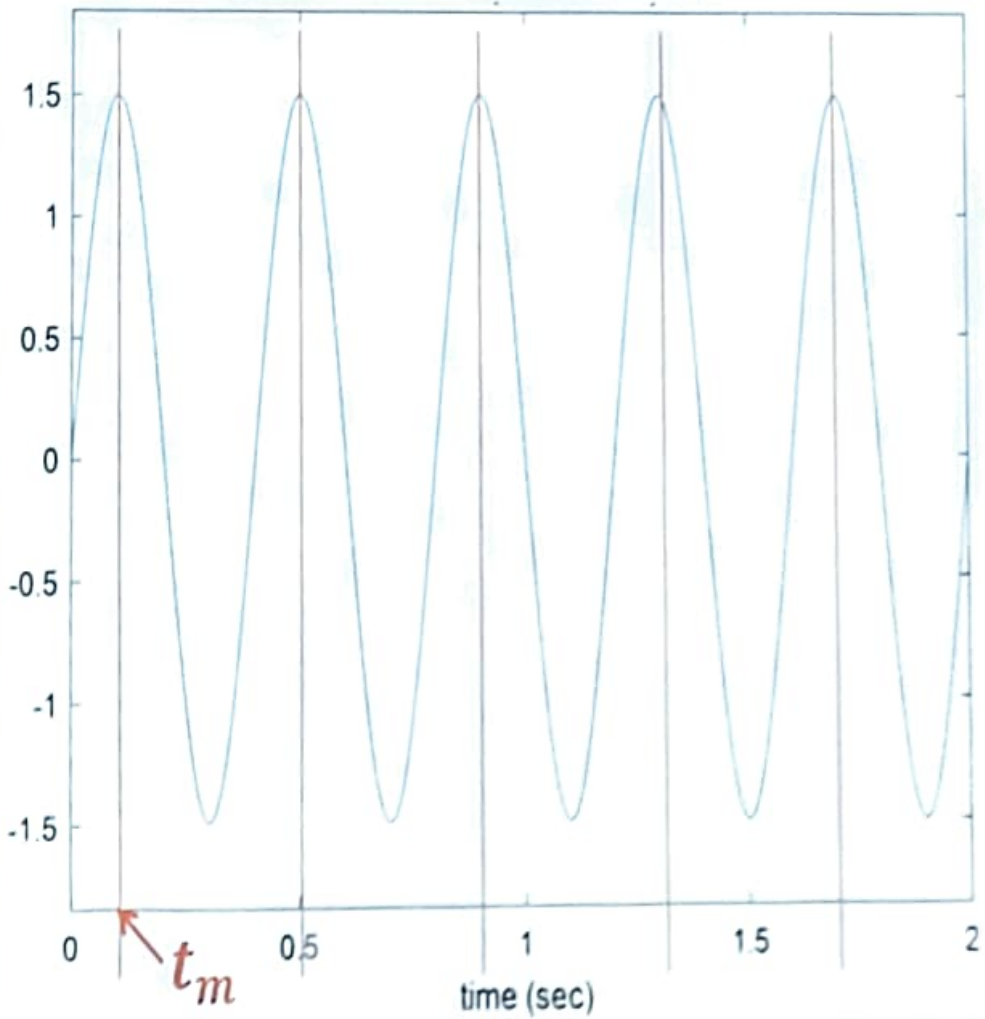
$T=0.4$ sec

$f=1/T=2.5$ Hz

$\omega = 2\pi f = 5\pi$ rad/sec

$\Phi = ?$

$x(t) = A\cos(\omega t + \Phi)$



$A = 1.5$

$T = 0.4$ sec

$f = 1/T = 2.5$ Hz

$\omega = 2\pi f = 5\pi$ rad/sec

$\Phi = -\omega t_m$

$= -(5\pi)*0.1$

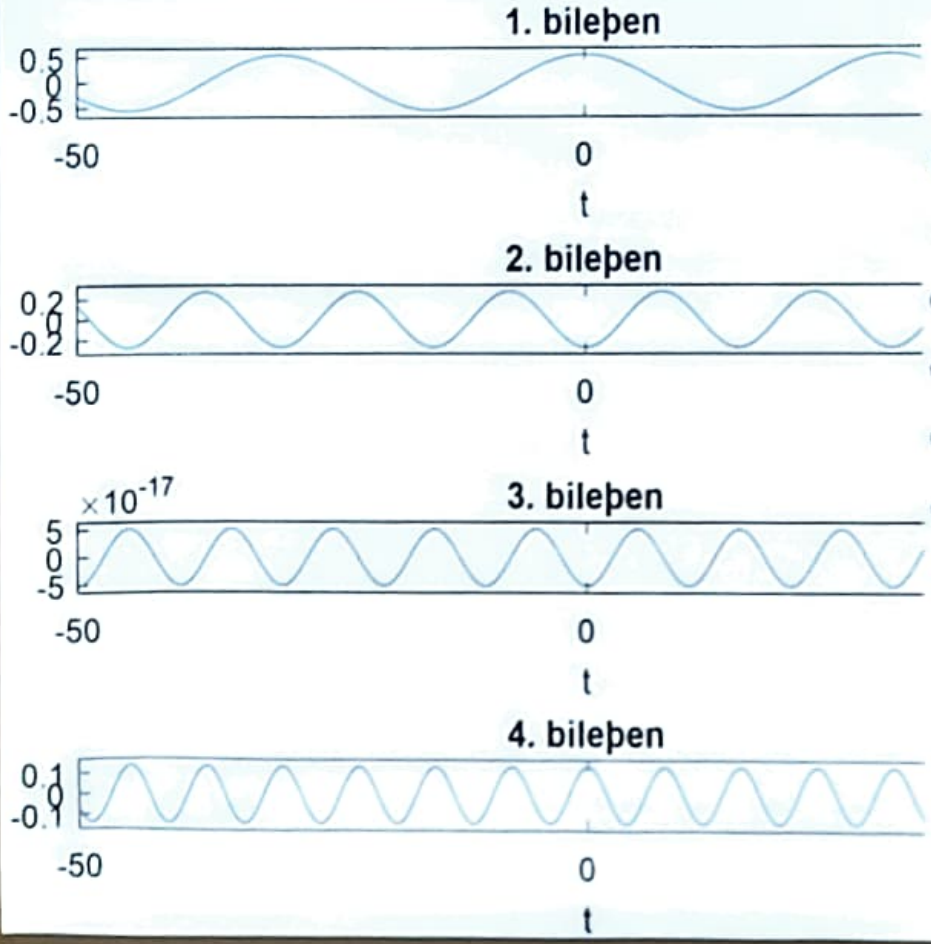
$= -0.5\pi$ rad

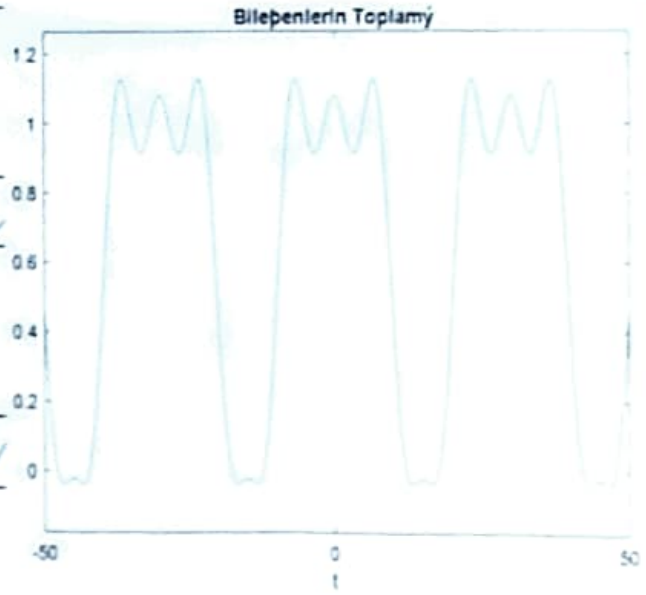
$x(t) = 1.5\cos(5\pi t - 0.5\pi)$

Harmonic Signal

- Signals with HARMONIC Frequencies
  - Add Sinusoids with $f_k = kf_0$

$$x(t) = A_0 + \sum_{k=1}^{N} A_k \cos(2\pi k f_0 t + \varphi_k)$$


1. bileþen
0.5
0
-0.5
-50
0
t
2. bileþen
0.2
0
-0.2
-50
0
t
×10^-17
3. bileþen
5
0
-5
-50
0
t
4. bileþen
0.1
0
-0.1
-50
0
t


Bileþenlerin Toplamý
1.2
1
0.8
0.6
0.4
0.2
0
-50
0
50
t

Energy and Power of a Signal

The signal energy in the signal $x(t)$ is

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt .$$

The signal power in the signal $x(t)$ is

$$P = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, dt .$$

Example

Compute the signal energy and signal power for the discrete-time signal

$$x(n)=\left(\frac{1}{4}\right)^n u(n).$$

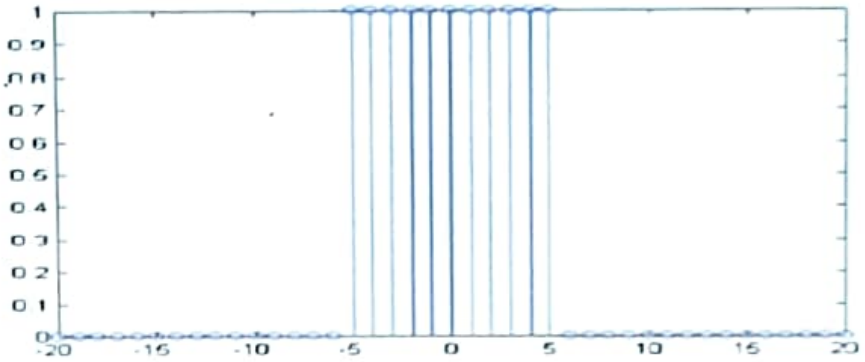
$$E=\sum_{n=-\infty}^{\infty}|x(n)|^2=\sum_{n=0}^{\infty}\left|\left(\frac{1}{4}\right)^n\right|^2=\sum_{n=0}^{\infty}\left(\frac{1}{16}\right)^n$$

The expression on the right hand side is a geometric series; hence, we have

$$E=\frac{1}{1-\frac{1}{16}}=\frac{16}{15}$$

Finite total energy signal

$$E_\infty < \infty, P_\infty = \lim_{T\to\infty} \frac{E_\infty}{2T} = 0$$

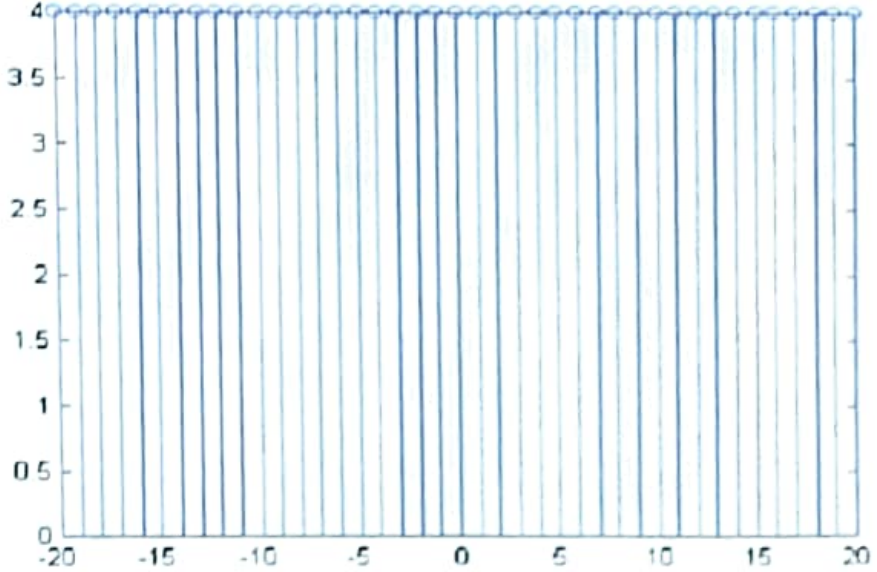


-5 < n <+5, x[n] = 1

otherwise x[n]=0. ENERGY =11.

Finite average power

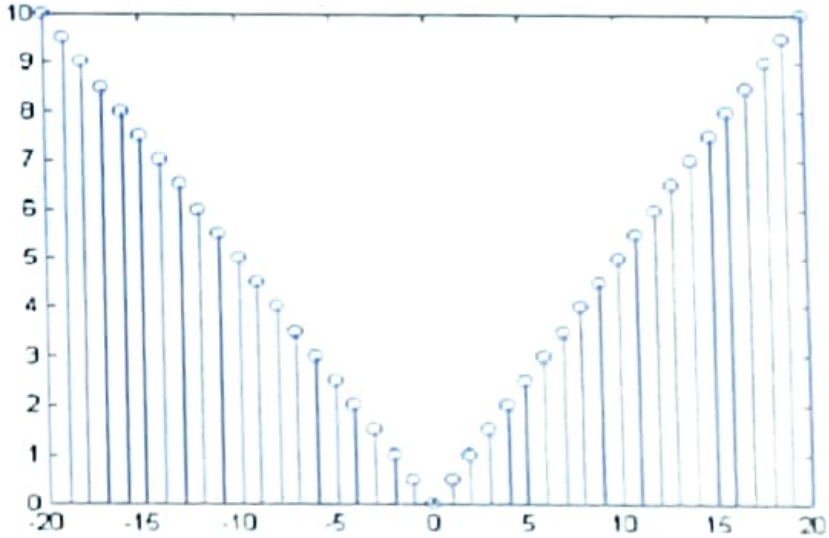
$$0 < P_\infty < \infty, \mathrm{E}_\infty = \infty$$



x[n] = 4, for all n.

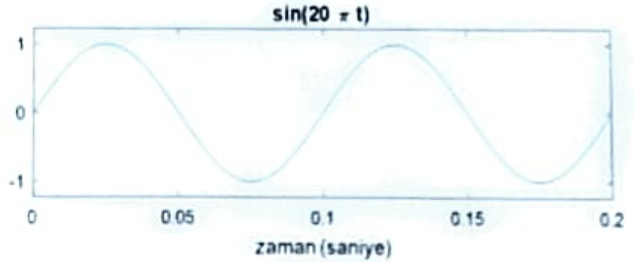
ENERGY =infinite, Power = 16

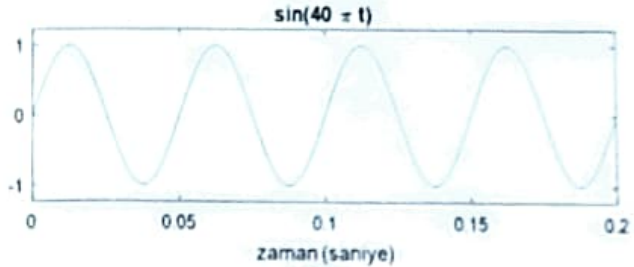
Neither power nor energy are finite $P_\infty = \infty, \mathrm{E}_\infty = \infty$

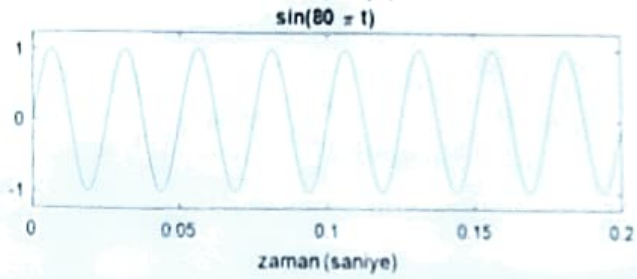


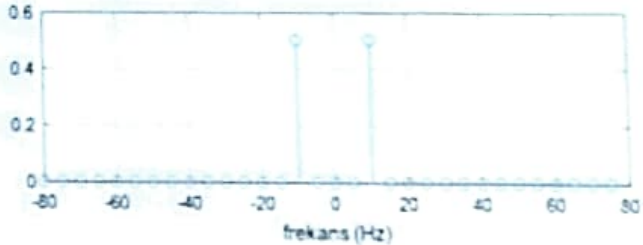
$$x[n] = 0.5|n|$$

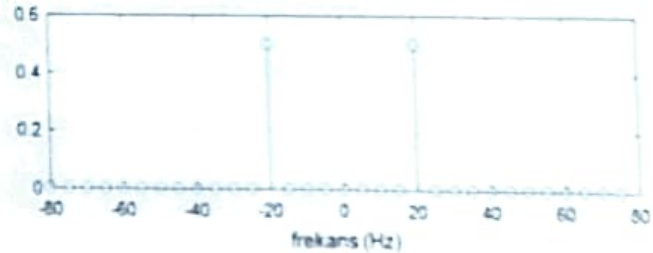
- Spectrum is the frequency domain representation of a signal

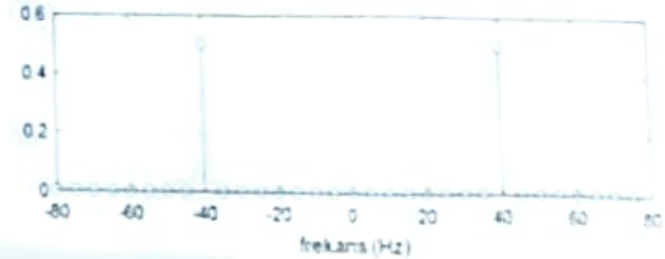




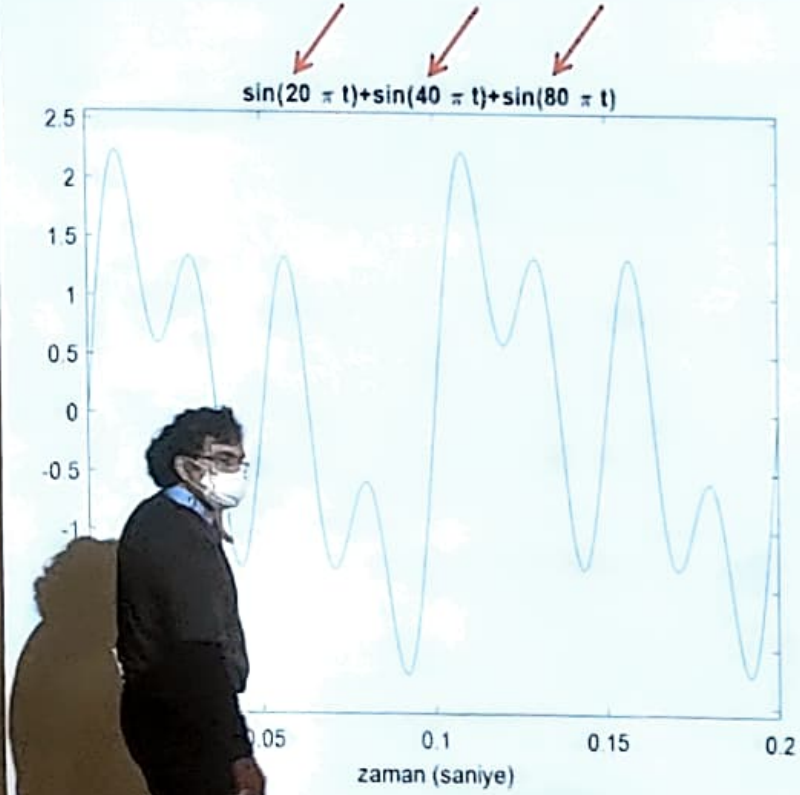


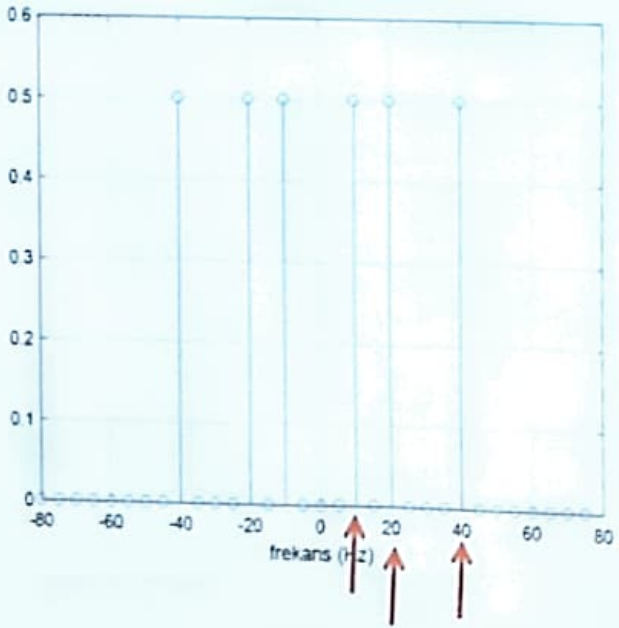






- Spectrum is the frequency domain representation of a signal





- Spectrum is the frequency domain representation of a signal

